United International **University** (**UIU**)

Dept. of Computer Science and Engineering (CSE)

Final Exam Year: **2025** Semester: **Fall**

Course: **CSE 3711** Title: **Computer Networks (Section – ALL)**

Marks: **40** Time: **2 Hours**

[Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.]

There are **Three (03)** questions. Answer **all of them**. All questions are of values indicated on the right-hand margin.

IMPORTANT NOTE: Candidates are required to answer the questions in **sequential order** (Q1, Q2 and Q3).

Q1. a) State **two key differences** between **Selective Repeat** and **Go-Back-N** reliability protocols. **[2]**

b) List **2 examples of features** provided by **TCP** that are not provided by **UDP**. **[2]**

c) **Explain** how **multiplexing** and **demultiplexing** occur at the Transport layer with **an example scenario**. **[3]**

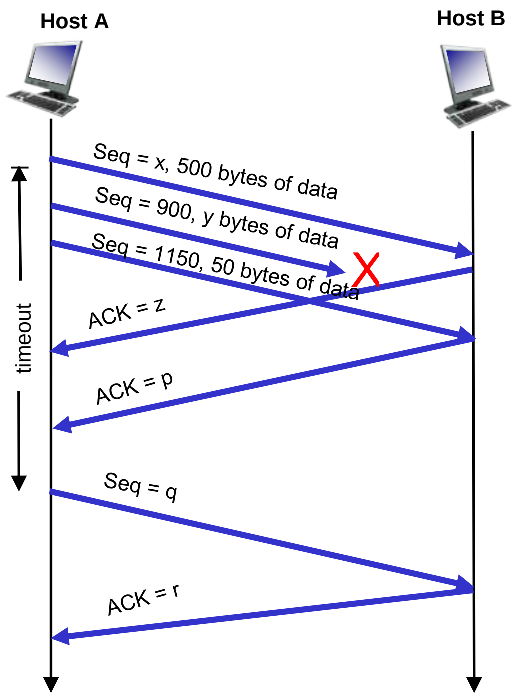
d) List the **following events** in the **order they occur** as **host A** opens a **TCP connection** to **host B**, transmits data and then **closes the connection**. Also, specify the sender and the receiver (Host A or Host B) for each event. **[3]**

(i) Send an ACK segment (ii) Close the connection (iii) Send a FIN segment (iv) Send a SYN segment

(v) Send a SYN-ACK segment (vi) Do the rest of the data exchange (vii) Send a FIN segment

(viii) Close the connection

e) Consider the **following diagram** that shows data transfer using TCP. Now, answer the **questions i, ii & iii**:



i. **Find** the values of **x, y, z, p, q & r**. **[3]**

ii. Suppose, the **first segment (Seq # x)** sent by host A to host B has **source port # 7000** & **destination port # 80**. For the **second segment** sent from Host A to B, what are the **source** and **destination port number**? **[1]**

iii. **List source and destination port numbers** for all three segments/ACKs sent by host B to host A. **[1]**

iv. Suppose, **segment 2** is **not lost**, what will be the **value of q** and **r** in this case? **[2]**

Q2. a) **Explain** why **routing** is a **global function**, while **forwarding** is a **local function** in the network layer. **[2]**

b) As shown in **Figure 1**, given **graph G = (V, E)**, where V is the set of routers and E is the set of links. Using **Dijkstra's link-state** routing algorithm, **compute** the least cost path from **node B** to all other nodes and **show** the resulting **least-cost-path tree/forwarding table** for B. **[5]**

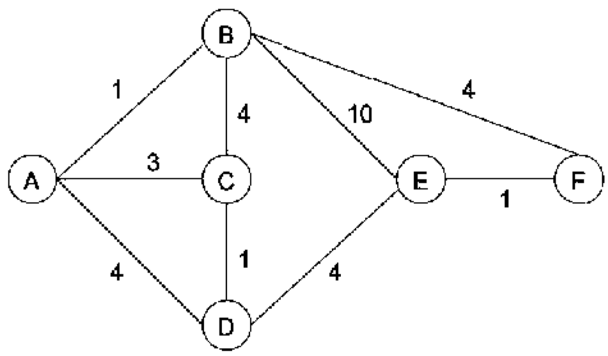


Figure 1: *Graph for Q.2(b)*

Q2. c) Consider the following network topology **(Figure 2)**, where Router A is configured to provide Network Address Translation (NAT) and Dynamic Host Configuration Protocol (DHCP) services.

i. **Identify** the IPv4 addressing **problems solved** by **Network Address Translation (NAT)**. **[2]**

ii. A **host** with **IP address 10.10.4.2** and **source port 6789** sends an **HTTPS request** (destination port **443**) to **google.com** (**142.250.75.142**). **Show** the **4 (four) steps** of the NAT process, clearly specifying the **source and destination IP addresses and port numbers** at each step with the aid of **a diagram**. **[3]**

iii. Now assume that, in addition to the communication shown in **part (ii)**, another host with **IP address 10.10.4.3** communicates simultaneously with:

- ✓ A web server at **100.10.10.11** using destination port **80** and client port **6789**, and
- ✓ An FTP server at **201.172.24.1** using destination port **21** and client port **9876**

Update the **existing NAT translation table** to include these new connections and **show** all entries. **[2]**

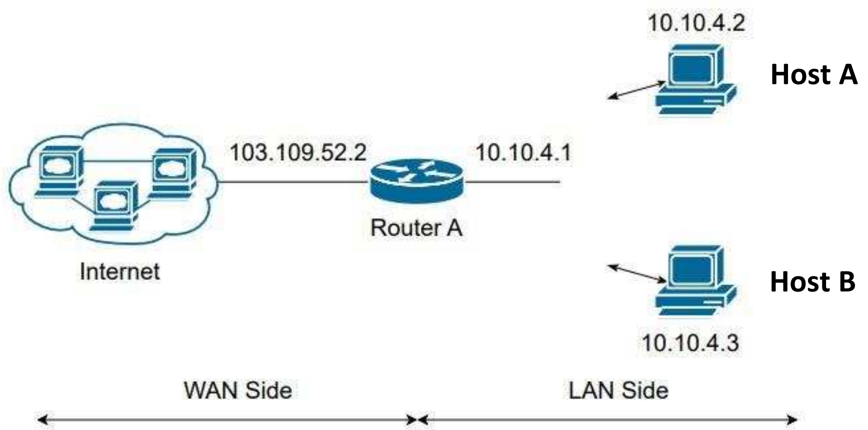


Figure 2: *Network Topology for Q.2(c)*

d) **State** the main functions of DHCP and **explain** the **4 (four) steps** of the **DHCP address allocation process** with the help of **a diagram** when a client joins a LAN. **[1 + 2 = 3]**

Q3. Consider **Figure 2** in **Q.2(c)** again. Answer the following questions assuming the **ARP table** of **Host A** given as follows:

IP Address	MAC Address
10.10.4.1	AA-AA-AA-AA-AA-AA
10.10.4.3	BB-BB-BB-BB-BB-BB

i. What are the **purpose** of **ARP**? **[1]**

ii. What is the **destination MAC address** in an **ARP query**? **[1]**

iii. **Host A** sends a packet to **Host B**. **Which MAC address** is used and **<u>why</u>**? **[2]**

iv. Now, **Host A** sends a packet to **8.8.8.8**. **Which MAC address** is used and **<u>why</u>**? **[2]**

<u>END OF THE QUESTION</u>